KAPASA MAKASA UNIVERSITY

SCHOOL OF APPLIED SCIENCE AND OPEN LEARNING

Department of Information and Communication Technology

FINAL EXAM

Program: Bachelor of Science in Cyber Security

Course Title: Mobile Device and Security and Forensic

Course Code: CYS 425

Date: 13TH November 2025

Duration: 3HRS

Total Marks: 100

General Instructions to Candidates:

1. Answer questions 1 and 2 and any other 3 three questions.
2. No unauthorized materials are allowed
3. No electronic devices allowed unless specified.
4. Start each answer on a new page.
5. Write clearly and neatly.

Please Note: *Students that are caught with foreign material, copying or behave in any manner that is viewed as examination malpractice/misconduct will be subjected to a disciplinary hearing.*

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[QUESTION 1]

You are a digital forensics expert with the Zambia Police Service's Cybercrime Unit. You have been called to a fraud investigation where the suspect's smartphone is a key piece of evidence. The suspect allegedly used the device to communicate with accomplices and transfer illicit funds.

(a) List FOUR different types of data commonly found on a mobile device that are crucial for a forensic investigation. (4 Marks)

(b) Explain TWO specific challenges you might face when attempting to extract data from this smartphone, relating to legal or ethical aspects. (4 Marks)

(c) Describe the first THREE critical steps you would take immediately upon receiving the mobile device to preserve the integrity of the evidence, starting from physical seizure. (6 Marks)

(d) The suspect's lawyer argues that extracting data from the cloud (e.g. Google Drive backups) violates their client's privacy beyond the scope of the warrant for the physical device.

i. Analyze the validity of this challenge. (3 Marks)

ii. Create a checklist of three points you would provide to the legal team to justify the need for cloud data acquisition in this investigation. (3 Marks)

[QUESTION 2]

A suspect in a fraud case uses a high-end Android phone with the latest operating system. Initial attempts at a logical acquisition have failed due to the device's strong security features.

(a) What is the primary purpose of the Secure Boot process in a modern mobile device?

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(2 Marks)

(b) Name two encryption mechanisms used in mobile devices. (2 Marks)

(c) Explain the legal implications of extracting data from a mobile device without consent or warrant. (2 Marks)

(d) Differentiate between Full Disk Encryption (FDE) and File-Level Encryption. (4 Marks)

(e) Outline the key components of a forensic report for a mobile device investigation. Include examples of the information required in each section. (10 Marks)

[QUESTION 3]

You are presenting your findings from a mobile forensics investigation in a high-profile court case. The defense counsel is challenging the integrity of your evidence, claiming the chain of custody was broken when the device was transported from the police station to your lab.

(a) What is the fundamental purpose of maintaining a Chain of Custody in a digital forensic investigation? (2 Marks)

(b) Explain the importance of hashing (using algorithms like SHA-256) in the evidence handling process. What does it prove about the data you have extracted? (4 Marks)

(c) During your testimony, you are asked what specific steps you took to minimize interaction with the device upon receipt. Describe THREE forensic best practices you followed from the moment you received the evidence bag. (6 Marks)

(d) The judge asks you to demonstrate that the evidence is authentic and reliable.

i. Evaluate the potential consequences of not having a properly documented chain of custody form. (4 Marks)

ii. One of the defense lawyers powers on a seized phone to "see if it works." Critique this

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action by explaining TWO specific types of evidence that could be altered or destroyed by this mistake. (4 Marks)

ii. Create a simple, 5-step procedural checklist for field agents to follow when seizing a mobile device, ensuring evidence integrity from the moment of seizure. (4 Marks)

[QUESTION 4].

Your digital forensics unit has a limited budget. You must decide between investing in a new commercial tool like Cellebrite UFED or relying on a suite of open-source tools like Autopsy.

(a) Define Mobile device forensics and explain its significance in criminal investigations. (4 Marks)

(b) Name ONE widely used commercial mobile forensic tool and ONE popular open-source alternative. (2 Marks)

(c) Compare open-source and commercial forensic tools by listing TWO advantages of each. (4 Marks)

(d) A specific case requires the extraction of data from a rare Chinese-manufactured Android phone.

i. Which type of tool (Commercial or Open-Source) would likely offer better support for this device, and why? (3 Marks)

ii. Regardless of your choice, what is one crucial step you must take to validate the findings from your forensic tool? (3 Marks)

(e) The head of your unit asks for a justification for the high cost of a commercial tool license.

i. Evaluate the situation by arguing FOR the purchase of the commercial tool, focusing on two key benefits that would justify the expense for a law enforcement context. (4 Marks)

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[QUESTION 5].

In a child exploitation case, the suspect's iPhone was seized, but critical evidence is believed to be stored in their iCloud account and synced with an Apple Watch.

(a) What are TWO major challenges forensic investigators face when dealing with evidence stored in the cloud? (2 Marks)

(b) Explain the concept of data jurisdiction and why it complicates cloud forensics investigations. (4 Marks)

(c) Describe the process of legally obtaining data from the suspect's iCloud account. (4 Marks)

(d) The suspect's Apple Watch is a potential source of evidence.

i. Analyze the specific types of forensic data you might extract from the Apple Watch (an IoT device) that could corroborate or expand upon evidence from the iPhone. (6 Marks)

ii. Create a checklist of four questions you would need to answer before beginning the acquisition of data from the Apple Watch. (4 Marks)

[QUESTION 6].

You are a forensic expert hired by a defense attorney. The prosecution's case relies heavily on evidence extracted from the defendant's phone. You are reviewing the methods used by the police forensics team.

(a) The forensic report states the tool used was "Cellebrite UFED." Why is it not sufficient for the report to merely state the tool used? What additional information about the process is required to establish authenticity? (4 Marks)

(b) The defendant claims they did not give consent for their phone to be searched, and there is no search warrant in the evidence file.

i. What legal principle does this potentially violate? (3 Marks)

ii. What would be the likely consequence for the evidence in court? (3 Marks)

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(c) During your review, you discover that the police examiner accessed personal photos and messages that were unrelated to the case.

i. Evaluate this action from an ethical perspective, referring to the principle of minimization. (5 Marks)

ii. Create a "Code of Conduct" for mobile forensic examiners that would help prevent such ethical breaches. (5 Marks)

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